

YINING HAN

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EDUCATION

University of Pennsylvania

Master of Applied Economics and Data Science (MAEDS)

Philadelphia, PA

Aug 2026 - Fall 2027

(Expected)

New York University Shanghai

B.S. in Mathematics | GPA: 3.7/4.0 | Dean's List (2024, 2025)

Shanghai, China

Sep 2022 - Jun 2026

- Relevant coursework: Machine Learning, Mathematical Statistics, Probability & Statistics, Data Structures, Basic Algorithms, Honors Analysis, Honors Linear Algebra, Honors Ordinary Differential Equations, and Mathematical Modeling.

EXPERIENCE

Zhejiang Dahua Technology Co., Ltd. - Data Research Institute

China

Data & AI Engineering Intern

May 2026 - Jul 2026

- Built an agentic synthetic-data generation pipeline from scratch using Claude Code with GLM and DeepSeek models, SQL, Git, Excel input/output utilities, and database scripts; the agent interpreted table schemas, field definitions, sample queries, and agent instruction files, then generated executable data-construction and SQL-ingestion code.
- Generated and populated a **multilingual test database containing 10M+ rows overnight** with Chinese, English, and Arabic records for large-scale functional and stress testing of a conversational Text-to-SQL / database question-answering agent.
- Developed LLM-based evaluation-corpus agents by adapting an existing Qinhuangdao/China workflow to a Qatar-specific international setting; produced approximately **2,000-3,000 Chinese and English test questions** for downstream system evaluation.
- Designed corpora covering zero-to-one query generation, one-to-three semantic generalization/paraphrasing, single-table questions, multi-table relational questions, and multi-turn drill-down/follow-up scenarios; automatically annotated each query with the database tables required to answer it.
- Conditioned synthetic-data generation on both database schemas and the generated evaluation queries, inspected the SQL catalog to verify fields and relationships, and repaired inconsistent shared-key generation across linked tables so records satisfied query semantics and cross-table constraints.
- Created SQL insertion and throughput-monitoring scripts, diagnosed locking/contention under a multithreaded approach, and revised the ingestion strategy to improve large-scale generation stability and efficiency; completed final validation in SQL and Navicat.
- Reviewed BIRD/Text-to-SQL research and schema-linking techniques, performed corpus and data quality assurance plus product functionality testing, and delivered artifacts that entered the team's formal test workflow and were later reused by teammates.

Zhejiang Feiluo Asset Management Co., Ltd.

China

Intern

Jul 2024 - Aug 2024

- Cleaned historical stock-price and trading-volume data and labeled samples for quantitative testing.
- Assisted with preliminary backtests of simple momentum factors, recorded results, and organized test outputs into standardized reports.
- Organized and archived more than 30 research memos and strategy documents, building familiarity with quantitative terminology and research workflows.

Haitong Securities (Zhejiang) Co., Ltd. - Binhang Branch

China

Investment Banking Intern, Debt Financing

Jul 2023 - Aug 2023

- Collected and organized market data for technology and consumer sectors and performed preliminary screening and classification of financial statements and public information from listed companies.
- Supported weekly industry reports with basic data visualization and charts covering industry trends and company-performance comparisons.

HAND Enterprise Solutions Co., Ltd.

China

Intern

Dec 2023 - Feb 2024

- Assisted with system configuration and data-collection tasks, supporting the technical setup and information-gathering phases of project cycles.
- Gained hands-on experience with IT service processes in an international business context and contributed to the department's operational workflow.

RESEARCH

A Comparative Study of Stock Daily and High-Frequency Forecasting Based on FinRL

Research Project

Jun 2025 - Sep 2025

- Constructed and trained FinRL models using daily and minute-level stock data and compared predictive accuracy, stability, and computational efficiency to evaluate whether high-frequency data provided a forecasting advantage.
- Performed data preprocessing, model training, and result analysis across multiple experiments.
- Built reinforcement-learning environments with Python and FinRL and debugged input formats across data frequencies.
- Visualized discrepancies between predictions and realized market performance and introduced regularization methods to mitigate overfitting.

COURSE PROJECTS

Music Genre Classification System Using Machine Learning

Course Project

Apr 2025 - May 2025

- Built a machine-learning model to classify music genres from audio features such as tempo, energy, and acousticness in a dataset of 50,000 Spotify songs.
- Designed a preprocessing pipeline with median imputation, one-hot encoding of categorical variables, label encoding of genre labels, and standardization of numerical features.
- Implemented and compared Random Forest, SVM, Gradient Boosting, and MLP classifiers with PCA and LDA dimensionality reduction.
- Achieved a **0.9261 weighted AUC** using a tuned MLP on scaled features, demonstrating strong discrimination across ten genres.
- Visualized song clusters in reduced-dimensional space and analyzed feature importance to interpret model behavior.
- Documented the modeling, evaluation, and visualization process in a technical report.

Chess Decision Tree with Alpha-Beta Pruning

Course Project

Apr 2024 - May 2024

- Developed a chess decision-making system that accepted FEN-formatted board states, searched efficiently, and returned optimal move strategies.
- Parsed FEN notation and generated legal moves with python-chess; implemented recursive Alpha-Beta search, minimax decision-making, pruning logic, a static evaluation function, and tree-state backtracking.
- Wrote project documentation and supported final system integration and testing.

Online Gomoku Game Based on a Distributed Chat System

Course Project

Nov 2023 - Dec 2023

- Designed and implemented a stable, low-latency online Gomoku server that processed player moves, synchronized board state in real time, determined game outcomes, and communicated with a GUI frontend.
- Extended an existing TCP protocol for game commands, created game-state management and win/loss logic, and synchronized data among multiple clients.
- Wrote server-side Python code using multithreading to process concurrent requests and maintain smooth gameplay.

LEADERSHIP & ACTIVITIES

L'Oreal BRANDSTORM 2023

Team Leader

Jan 2023 - Mar 2023

- Led a team in designing two beauty-product or technology concepts integrating new media with diversity and inclusion.
- Organized the video and slide presentation of the concepts and advanced to the semifinals.

NYU SHDIC 2023

Team Leader

Oct 2023 - Nov 2023

- Managed tasks, promoted ideas, and motivated teammates to develop an AI-based solution for accessible reading and movie-watching experiences for visually impaired users.
- Strengthened team management, collaboration, and promotion experience while exploring inclusive technology design.

Dean's Undergraduate Research Fund 2024

Team Leader

Mar 2024 - May 2024

- Led research-question formulation, survey and interview design, data collection, and experimental-report writing.
- Coordinated schedules and interviewees while developing communication, teamwork, and data-analysis skills.

TECHNICAL SKILLS

Programming & Data: Python, SQL, C, C++, Git, Excel, SPSS, EViews

AI / ML: Machine Learning, Reinforcement Learning, LLM Agents, Synthetic Data Generation, Text-to-SQL, Prompt Engineering

Tools & Platforms: Claude Code, GLM, DeepSeek, FinRL, Navicat, Microsoft Office